

SMART CANE

OBSTACLE DETECTION

Embedded Systems Project

By:

Youssef Rami

Nour Tamer

Mariam ElGhobary

PROBLEM

White canes used by visually impaired members of society detect ground-level obstacles, but nothing above.

**Consequent hazards of indoor navigation:
Objects such as :**

- tables, shelves, open cabinets and other obstacles at body height remain completely undetected.
- Only one direction can be explored at a time. (the direction the cane is directed towards)



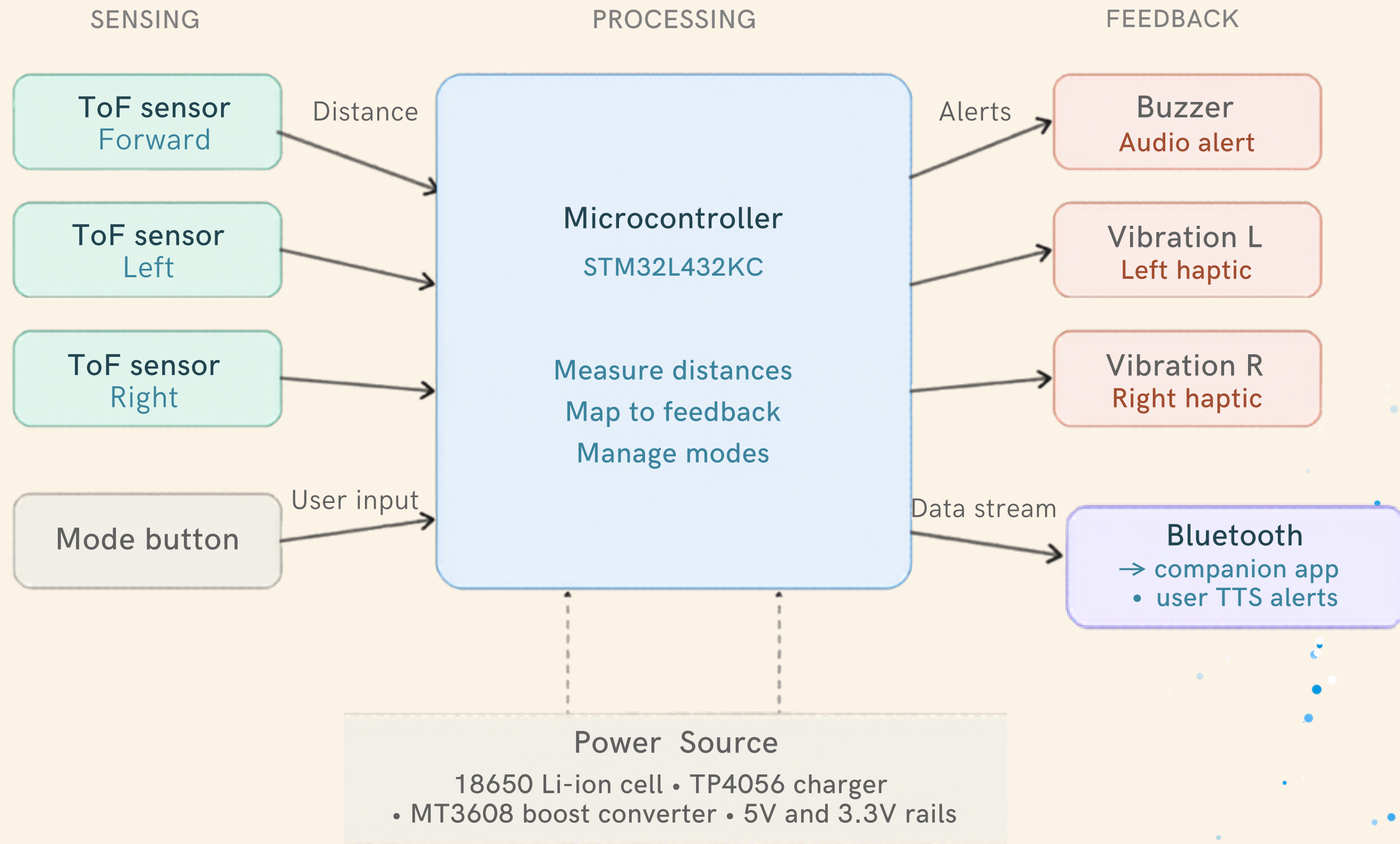
SOLUTION

We aim to create a clip-on attachment for the upper grip of the standard white cane, extending its capability beyond the ground to detect obstacles at body height in real time. Four feedback modes (buzzer-only, vibration-only, combined, neither/silent) let the user adapt to any environment.

HOW IT WORKS

- 3 VL53L1X laser Time-of-Flight sensors continuously scan for obstacles at body height (forward, left, and right)
- Audio feedback via distinct beep patterns per direction, with beep speed indicating distance + companion app converting obstacle data to spoken alerts via Text-to-Speech through the user's own earphones or headset
- Bluetooth streams obstacle data to companion app for TTS alerts
- Fully integrated 18650 rechargeable battery providing 8–12 hours of continuous use

BLOCK DIAGRAM



Main COMPONENTS

Component	Quantity
NUCLEO-L432KC	1
VL53L1X ToF Sensor	3
HC-05 Bluetooth Module	1
Buzzer	1
Coin Vibration Motor	2
Push Button	1
18650 Li-ion Cell	1
TP4056 Charging Module	1
MT3608 Boost Converter	1
Slide Switch	1

FUNCTIONAL REQUIREMENTS

- 1. Detect obstacles within 2m (forward, left, and right)**
- 2. Deliver distinct beep patterns per direction, speed increasing with proximity alongside TTS alerts through companion app when phone is paired**
- 3. Support four feedback modes switchable via push button**
- 4. Transmit live distance data to a paired phone via Bluetooth**
- 5. Poll all three sensors without interference**
- 6. Response time under 500ms at walking pace**

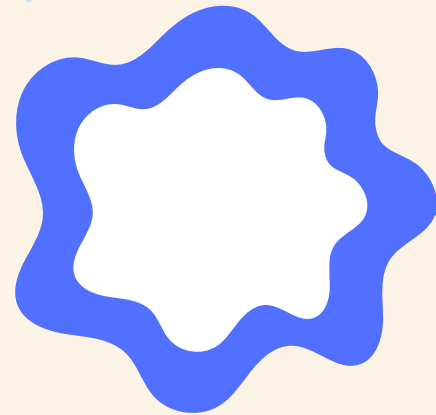
CHALLENGES

- 1. VL53L1X is relatively expensive**
- 2. Sensor range decreases in direct sunlight (still within 2 metres) and is less accurate with reflective materials**
- 3. Distinguishable feedback such that beep patterns must be distinct enough for a user to reliably tell directions apart**
- 4. Power constraints: device must run long enough on a portable power source to be practically usable**



PLAN

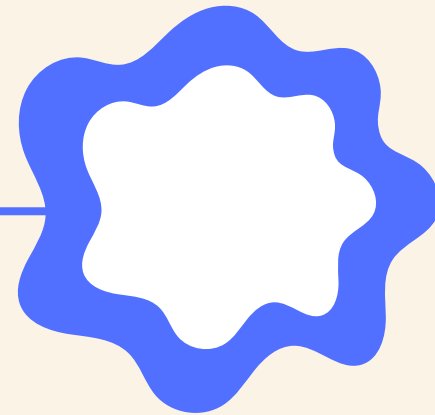
Week 1



Setup & Individual Subsystems

Single sensor test,
HC-05 SPP link,
power path
validation

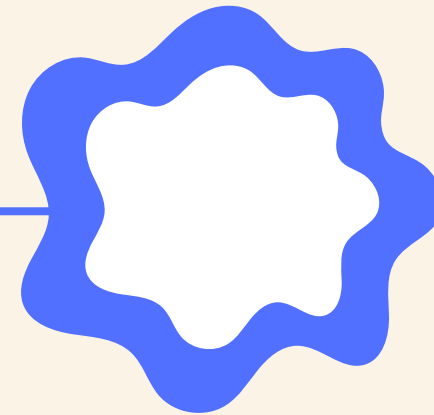
Week 2



Sensor Integration

Three sensors on
I2C bus, buzzer +
vibration PWM,
start companion
app

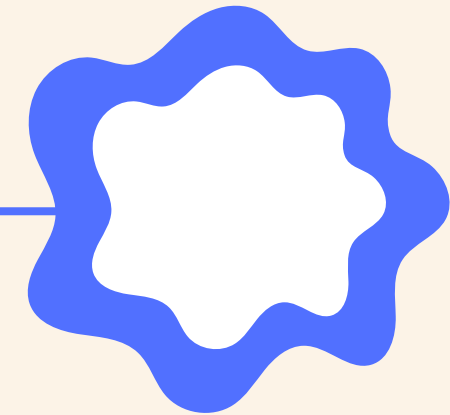
Week 3



Full System Integration

Full system
integration, mode
switching, TTS
alerts connected

Week 4



Testing & Demo

Tune thresholds, fix
bugs, prepare
demo and finalize
wiki

THANK YOU

